

Topics to discuss

- Romberg Integration
- Numerical Problem

Romberg Integration :

Suppose we have to compute,

$$I = \int_a^b f(x) dx$$

If we apply trapezoidal rule once, we get an approximation.
If we then divide the interval into more subintervals,
trapezoidal rule becomes better.

Romberg Integration takes these trapezoidal approximations
and then removes much of the error systematically.

It means that,

Romberg = Repeated trapezoidal rule + error cancellation

for one interval means, $n=1$

$$y = \int_{x_0}^{x_1} f(x) dx$$

$$h = \frac{x_1 - x_0}{1} = x_1 - x_0$$

$$\therefore T_1 = \frac{x_1 - x_0}{2} [y_0 + y_1]$$

for n equal subintervals,

$$T_n = \frac{h}{2} \left[y_0 + 2 \sum_{i=1}^{n-1} y_i + y_n \right]$$

Now in Romberg, we usually take:

$n=1$

then $n=2$

then $n=4$

then $n=8$

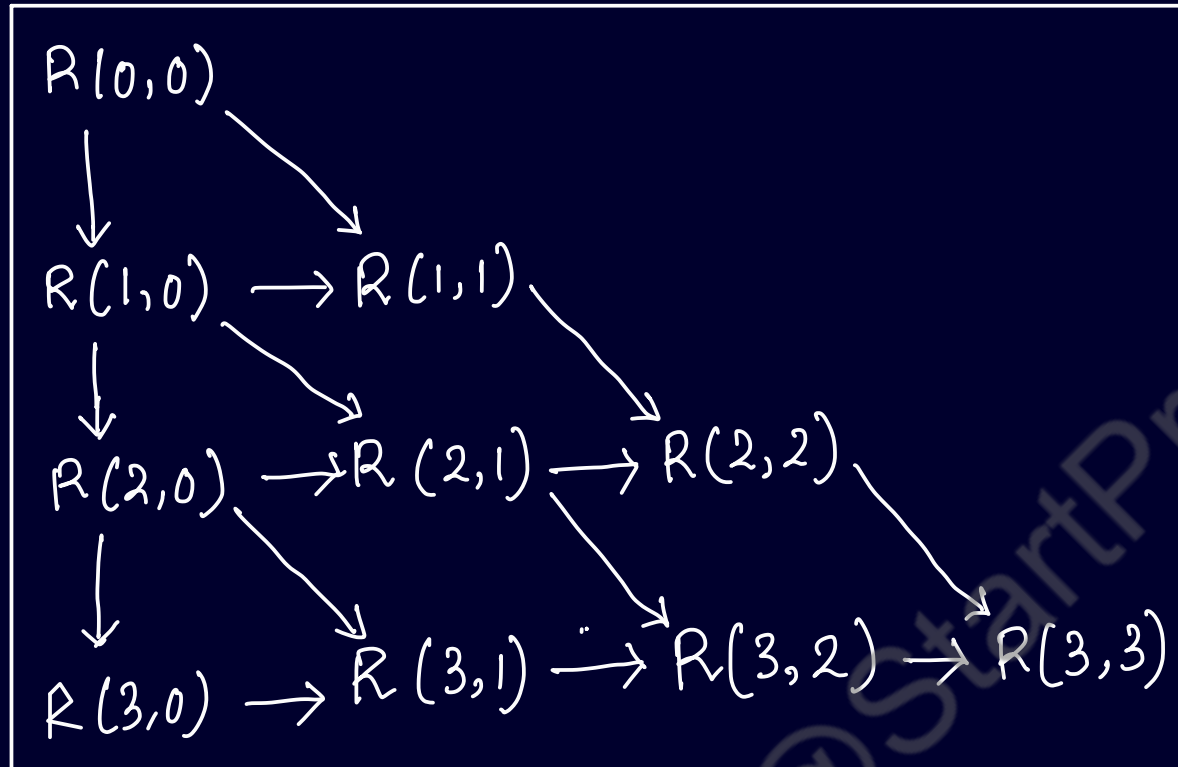
then $n=16$

So each time the step size is halved.

$$R_{ij} = \frac{4^j R_{i,j-1} - R_{i-1,j-1}}{4^j - 1}$$

This is known as Romberg Integration formula.
If we expand this equation, we will get lower-diagonal matrix.

The elements of the matrix R are computed row by row in the order indicated in the next Figure.



Elements in the first column represent trapezoidal Rule at $h, \frac{h}{2}, \frac{h}{4}, \frac{h}{8}, \frac{h}{16}$ etc.

They can be evaluated recursively as follows:

$$h = b - a$$

$$R(0,0) = \frac{h}{2} [y_0 + y_1]$$

$$R(i,0) = \frac{R(i-1,0)}{2} + h_i \sum_{k=1}^{2^{i-1}} y_{2k-1}$$

for, $i \geq 1, 2, \dots$

$$\text{where, } h_i = \frac{b-a}{2^i}$$

$$x_k = a + k h_i$$

This equation is known as recursive trapezoidal rule.

In Simple term,

* To find first Romberg correction, means for 2nd column, use

$$\frac{4B-A}{3} ; A \text{ \& B are two successive rows of 1st column.}$$

* for 3rd column, $\frac{16B-A}{15}$; A & B are two successive rows of 2nd column.

* for 4th column, $\frac{64B-A}{63}$; A & B are two successive rows of 3rd column.

* for 5th column, $\frac{256B-A}{255}$; A & B are two successive rows of 4th column.

Numerical Problem :

Q: Compute Romberg estimate upto R_{22} for

$$\int_0^1 \frac{1}{1+x} dx$$

solution: We have to find $R_{00}, R_{10}, R_{20}, R_{11}, R_{21}, R_{22}$

Romberg table starts from trapezoidal values,

and the higher values are:

$$R_{kj} = R_{k,j-1} + \frac{R_{k,j-1} - R_{k-1,j-1}}{4^{j-1} - 1}$$

First trapezoidal value R_{00}, T_1

Here no. of subintervals $n=1$

x	0	1
$f(x)$	1	0.5

$$\therefore h = \frac{1-0}{1} = 1$$

$$f(x) = \frac{1}{1+x}$$

$$T_1 = \frac{h}{2} [1 + 0.5] = \frac{1}{2} \times 1.5 = 0.75$$

Second trapezoidal value, R_{10}, T_2

$$n=2, \quad h = \frac{1-0}{2} = 0.5$$

x	0	0.5	1
$f(x)$	1	0.667	0.5

$$\begin{aligned} \therefore T_2 &= \frac{h}{2} [1 + 0.5 + 2 \times 0.667] \\ &= 0.708335 \end{aligned}$$

Third trapezoidal Rule, R_{20}

$$n = 4 \quad \therefore h = \frac{1-0}{4} = 0.25$$

x	0	0.25	0.5	0.75	1
$f(x)$	1	0.80	0.667	0.571	0.5

$$\therefore T_3 = \frac{h}{2} [1 + 0.5 + 2(0.80 + 0.667 + 0.571)]$$

$$= 0.6970$$

Computing, R_{11} using, $\frac{4B-A}{3}$

$$= \frac{4 \times 0.708335 - 0.75}{3}$$

$$= 0.694444$$

$$R_{00} = 0.75$$

$$R_{10} = 0.708335 \quad R_{11} = 0.694444$$

$$R_{20} = 0.6970 \quad R_{21} = 0.693221 \quad R_{22} = 0.693140$$

Computing R_{21} using $\frac{4B-A}{3}$

$$= \frac{4 \times 0.6970 - 0.708335}{3}$$

$$= 0.693221$$

Computing, R_{22} using $\frac{16B-A}{15}$

$$= \frac{16 \times 0.693221 - 0.69444}{15}$$

$$= 0.693140$$

So, the final answer is, $\int_0^1 \frac{1}{1+x} dx = 0.693140$
Ans.

Connect with me



[Start Practicing](#)



[i._am._arfin](#)



[Arfin Parween](#)



[Arfin Parween \(Subscribe here too\)](#)



[Arfin Parween](#)



[Arfin Parween](#)